

1. Which binary number causes overflow in a 4-bit 2's complement system?
 A) 0111 + 0001
 B) 0101 + 0010
 C) 1000 + 1000
 D) Both A and C
2. The excess-3 code of decimal number 596 is:
 A) 1000 1100 1111
 B) 1010 1100 1111
 C) 1000 1110 1001
 D) 1100 1010 1111
3. Which binary code is self-complementing?
 A) BCD
 B) Gray
 C) Excess-3
 D) ASCII
4. In Gray code, consecutive numbers differ by:
 A) One bit only
 B) Two bits
 C) MSB only
 D) LSB only
5. Which statement is TRUE about ASCII?
 A) It is a weighted code
 B) It supports only numbers
 C) It is a 7-bit character code
 D) It is self-complementing
6. The binary representation of -7 using 1's complement (4-bit) is:
 A) 0111
 B) 1001
 C) 0110
 D) 1110
7. Which code minimizes errors during mechanical position sensing?
 A) BCD
 B) Excess-3
 C) ASCII
 D) Gray code
8. The decimal value of binary number 11010.101 is:
 A) 26.625
 B) 26.5
 C) 27.25
 D) 25.625
9. Which OS is MOST suitable for real-time missile control systems?
 A) Time-sharing OS
 B) Distributed OS
 C) Hard Real-Time OS
 D) Batch OS
10. Which OS allows multiple users to work simultaneously on a single system?
 A) Single-user multitasking
 B) Multi-user OS
 C) Distributed OS
 D) Network OS
11. Which OS type focuses on resource sharing across networked machines?
 A) Network OS
 B) Distributed OS
 C) Batch OS
 D) Embedded OS
12. UNIX is an example of:
 A) Single-user single-task OS
 B) Multi-user multitasking OS
 C) Single-user multitasking OS
 D) Batch OS
13. Which OS executes jobs in groups without user interaction?
 A) Time-sharing
 B) Real-time
 C) Batch OS
 D) Distributed
14. Android OS is classified as:

- A) Embedded OS only
 - B) Network OS
 - C) Mobile OS based on Linux
 - D) Real-time OS
15. Which interface requires memorization of commands?
 - A) GUI
 - B) CUI
 - C) Touch UI
 - D) Menu-driven UI
 16. Windows 95 primarily introduced which UI feature?
 - A) CLI
 - B) Touch UI
 - C) GUI with multitasking
 - D) Distributed UI
 17. Single-user multitasking OS means:
 - A) Multiple users, one task
 - B) One user, multiple tasks
 - C) Multiple users, multiple tasks
 - D) One user, one task
 18. MS-DOS is an example of:
 - A) Multi-user multitasking
 - B) Single-user multitasking
 - C) Single-user single-tasking
 - D) Batch processing
 19. Which UI is MOST user-friendly for beginners?
 - A) CLI
 - B) CUI
 - C) GUI
 - D) Shell interface
 20. Which program is executed first during booting?
 - A) Operating System
 - B) POST
 - C) BIOS/UEFI
 - D) Boot loader
 21. POST stands for:
 - A) Power On Software Test
 - B) Program On Self Test
 - C) Power On Self Test
 - D) Primary OS Startup Test
 22. The boot loader is stored in:
 - A) RAM
 - B) Cache
 - C) Hard Disk MBR
 - D) CPU registers
 23. First generation computers used:
 - A) Transistors
 - B) ICs
 - C) Vacuum tubes
 - D) Microprocessors
 24. Which language was dominant in 2nd generation computers?
 - A) Machine language
 - B) Assembly language
 - C) Java
 - D) Python
 25. Microprocessors were introduced in which generation?
 - A) Second
 - B) Third
 - C) Fourth
 - D) Fifth
 26. AI-based systems are associated with:
 - A) Third generation
 - B) Fourth generation
 - C) Fifth generation
 - D) Second generation
 27. Which OS was the first commercially successful GUI-based OS?
 - A) UNIX
 - B) MS-DOS
 - C) Windows 3.1
 - D) macOS (System 1)
 28. Convert in gray code of 1 = ?
 29. Two Example of Analog Computer = ?
 30. Two Example of Digital Computer = ?